

A leading-term asymptotic for the perimeter deficit of randomized integer convex hulls

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Abstract

For a planar convex body with C^2 boundary and strictly positive curvature, we derive a leading-term asymptotic for the expected perimeter deficit of its randomized integer convex hull. Exact averaging over lattice translations reduces empty-cap probabilities to bounded continuous tests on the space of unimodular lattices. Expanding-circle equidistribution and an integrable uniform tail bound give the complete expectation. A stationary parabolic mass transport and primitive marked Haar unfolding identify the coefficient with an explicit row-model integral. Exact rational series evaluation yields $0.71923298 < C_{\text{void}} < 0.71923337$, hence the rounded value 0.719233. The previously established order $R^{-1/3}$ is credited separately from the limit and coefficient identification. No second-order term, effective convergence rate, flat-point extension, or scientific priority is claimed. The proofs and computations have been developed and checked with AI assistance and remain submitted for independent mathematical scrutiny.

Résumé. Pour un corps convexe plan à bord C^2 et à courbure strictement positive, nous établissons une asymptotique du déficit moyen de périmètre de sa coque entière randomisée. La convergence des probabilités de calottes vides, leur domination intégrable et un transport de masse dans le modèle parabolique identifient le coefficient principal. Un calcul rationnel certifie son encadrement. Le second terme, les points plats et un taux effectif restent ouverts. Les contributions des IA sont déclarées ; aucune relecture mathématique humaine indépendante n'est attestée.

1 Model, main result and relation to earlier work

Let $K \subset \mathbb{R}^2$ be a compact convex body with nonempty interior. Write $P(K)$ for its perimeter. For $R > 0$, set

$$L = \rho(\theta)(\mathbb{Z}^2 + t), \quad \theta \sim \text{Unif}[0, 2\pi), \quad t \sim \text{Unif}[0, 1)^2, \quad Q_R = \text{conv}(L \cap RK), \quad (1.1)$$

with independent θ, t , and $\rho(\theta)$ the Euclidean rotation. Translation is taken modulo the integer lattice *before* rotation. All limiting statements below concern a fixed body K .

Theorem 1. *Suppose ∂K is C^2 and its curvature κ is strictly positive everywhere. Then*

$$\mathbb{E}[P(RK) - P(Q_R)] = C_{\text{void}} \left(\int_{\partial K} \kappa^{4/3} ds \right) R^{-1/3} + o_K(R^{-1/3}). \quad (1.2)$$

The coefficient is independent of K and admits the identities

$$C_{\text{void}} = \int_0^\infty v_H(z) dz = \frac{1}{\zeta(2)} \int_0^\infty aG(a) da, \quad 0.71923298 < C_{\text{void}} < 0.71923337. \quad (1.3)$$

Here v_H is the Haar-affine avoidance probability of the parabolic cap defined in Section 2, and G is the row-model expectation defined in Section 8.

The randomized integer-hull model was studied by Bárány–Matoušek [4]. Ngoc–Reitzner [2] already prove the order $R^{-1/3}$ for the smooth planar case. Their published Section 5 asks for convergence of normalized empty-cap probabilities for ellipsoids; arXiv v1 formulates the question for smooth bodies. The present argument addresses this convergence and the coefficient identification in dimension two. Its uniform-tail method follows their Theorem 5.1, using the avoidance estimate attributed to Bárány [3], whose planar case goes back to [4]. These attributions are part of the result’s scope.

This manuscript consolidates the three proof notes and their regularity supplement [6, 7, 8, 9]. It develops the earlier candidate model [5]; that version’s historical conjectural claims remain attached to their original date. Literature priority beyond the references checked here is not established. The presentation supplies a mathematical derivation and a numerical certificate, while distinguishing both from AI or human review status.

2 Normalized caps and their uniform geometric limit

For $u_\varphi = (\cos \varphi, \sin \varphi)$, let $p_K(\varphi)$ be the support point, κ_φ its curvature and h_K the support function. Put

$$p_R(\varphi) = R p_K(\varphi), \quad q_{R,\varphi} = \kappa_\varphi / R, \quad h = q_{R,\varphi}^{1/3} z.$$

For $z \geq 0$, define the physical cap and its avoidance probability by

$$A_{R,\varphi}(z) = \{w \in RK : w \cdot u_\varphi \geq h_{RK}(u_\varphi) - h\}, \quad v_R(z, \varphi) = \mathbb{P}\{L \cap A_{R,\varphi}(z) = \emptyset\}. \quad (2.1)$$

Let μ_{aff} be Haar probability on affine unimodular lattices, and set

$$\mathcal{P} = \{(X, Y) : Y \geq X^2/2\}, \quad A(z) = \{X^2/2 \leq Y \leq z\}, \quad v_H(z) = \mu_{\text{aff}}\{\Lambda : \Lambda \cap A(z) = \emptyset\}. \quad (2.2)$$

Choose the positively oriented frame $(\tau_\varphi, -u_\varphi)$, where $\tau_\varphi = (-\sin \varphi, \cos \varphi)$. In coordinates tangent to K and measuring inward depth, the boundary is $y = f_\varphi(x)$ with

$$f_\varphi(0) = f'_\varphi(0) = 0, \quad f''_\varphi(0) = \kappa_\varphi.$$

There is a common graph interval. Indeed, an arclength parametrization γ is C^2 ; its tangential coordinate at $\gamma(s)$ has derivative 1 at s . Uniform continuity of the tangent gives a common interval where this derivative is at least $1/2$. Inverting this coordinate gives second graph derivatives jointly continuous in the support point and graph coordinate. Compactness therefore supplies a nondecreasing modulus $\omega_K(r) \rightarrow 0$ such that

$$\left| f_\varphi(x) - \frac{\kappa_\varphi x^2}{2} \right| \leq \frac{x^2}{2} \omega_K(|x|). \quad (2.3)$$

Also $0 < \kappa_- \leq \kappa_\varphi \leq \kappa_+ < \infty$.

The boundary of RK is $y = R f_\varphi(x/R)$. Apply the determinant-one map

$$X = q_{R,\varphi}^{1/3} x, \quad Y = q_{R,\varphi}^{-1/3} y. \quad (2.4)$$

Its normalized graph $g_{R,\varphi}$ satisfies, on each fixed bounded interval of X ,

$$\left| g_{R,\varphi}(X) - \frac{X^2}{2} \right| \leq \frac{X^2}{2\kappa_-} \omega_K(\kappa_-^{-1/3} R^{-2/3} |X|) = o(1), \quad (2.5)$$

uniformly in φ .

Let $B_{R,\varphi}(z)$ denote the centered, normalized cap. For every fixed $Z < \infty$, the physical depths in K are $O_K(R^{-4/3}Z)$. Uniqueness of support points and compactness confine these caps to the common graph neighborhoods. Shrinking them if needed gives $f_\varphi(x) \geq \kappa_\varphi x^2/4$, hence

$$B_{R,\varphi}(z) \subset \{|X| \leq 2\sqrt{Z}, 0 \leq Y \leq Z\} \quad (0 \leq z \leq Z)$$

for all sufficiently large R . Their vertical sections and those of $A(z)$ end at the same height z . The function $r \mapsto (z - r)_+$ is 1-Lipschitz, so

$$\sup_{\varphi, 0 \leq z \leq Z} |B_{R,\varphi}(z) \triangle A(z)| \leq C_{K,Z} \omega_K(C_{K,Z} R^{-2/3}) = o(1). \quad (2.6)$$

For C^3 boundaries the right-hand side can instead be bounded by $C_{K,Z} R^{-2/3}$. That geometric bound is not a rate for the full limit theorem.

3 Translation averages and expanding-circle equidistribution

We now use row vectors. Write $X_0 = \text{SL}(2, \mathbb{Z}) \backslash \text{SL}(2, \mathbb{R})$ with Haar probability μ_0 . For a bounded measurable set B and $M \in \text{SL}(2, \mathbb{R})$, let

$$F_B(M) = \int_{[0,1]^2} \mathbf{1}\{(\mathbb{Z}^2 + a)M \cap B = \emptyset\} da. \quad (3.1)$$

Integer changes of basis preserve torus measure, so F_B descends to X_0 .

Lemma 1. *For every measurable D of finite area and every unimodular M ,*

$$\int_{[0,1]^2} \#((\mathbb{Z}^2 + a)M \cap D) da = |D|.$$

Consequently $|F_B(M) - F_{B'}(M)| \leq |B \triangle B'|$. If B is compact with boundary of area zero, then $F_B \in C_b(X_0)$.

Proof. Tonelli unfolds the counting sum to $\int_{\mathbb{R}^2} \mathbf{1}_D(xM) dx = |D|$. A difference of avoidance indicators requires a point in the symmetric difference, giving the stated bound. For continuity, if $M_j \rightarrow M$, boundedness of B and of the inverse matrices leaves only finitely many candidate integer vectors uniformly in the torus translation. Almost every translation avoids ∂B by the intensity identity. Each candidate's membership then stabilizes. Dominated convergence proves continuity, which descends to the quotient. $\square$

Let O_φ have columns $\tau_\varphi, -u_\varphi$, and $D_q = \text{diag}(q^{1/3}, q^{-1/3})$. The normalized lattice is exactly

$$(\mathbb{Z}^2 + t)\rho(\theta)^T O_\varphi D_q - p_R O_\varphi D_q = (\mathbb{Z}^2 + a)M, \quad (3.2)$$

where $M = \rho(\theta)^T O_\varphi D_q$ and $a = t - p_R \rho(\theta) \pmod{\mathbb{Z}^2}$. Conditional on θ , a is precisely uniform on the torus. Thus the diverging center is absorbed without a Diophantine assumption.

Define

$$s_{R,\varphi} = \frac{1}{3} \log(R/\kappa_\varphi), \quad \Phi_s = \text{diag}(e^{-s}, e^s), \quad \Psi_s F = \frac{1}{2\pi} \int_0^{2\pi} F(\rho(\alpha)\Phi_s) d\alpha.$$

Since $\rho(\theta)^T O_\varphi$ is uniformly distributed on rotations, Lemma 1 and (2.6) give

$$\sup_{\varphi, 0 \leq z \leq Z} |v_R(z, \varphi) - \Psi_{s_{R,\varphi}} F_{A(z)}| = o(1). \quad (3.3)$$

The level-one, dimension-two specialization of Marklof–Strömbergsson’s Corollary 5.9 [1] yields

$$\Psi_s F \longrightarrow \int_{X_0} F d\mu_0 \quad (s \rightarrow \infty), \quad F \in C_b(X_0). \quad (3.4)$$

The angular measure is absolutely continuous and the rotation map has nonvanishing spherical derivative. The omitted endpoints of $(0, 2\pi)$ have measure zero. Their concluding remark in Section 5.4 states that the SL result follows from diagonal mixing without Ratner theory.

Disintegrating Haar-affine probability into base Haar and uniform torus translations gives

$$\int_{X_0} F_{A(z)} d\mu_0 = v_H(z).$$

The cap boundary has area zero, so the continuous-test condition is met. Furthermore $\inf_{\varphi} s_{R,\varphi} \rightarrow \infty$. Finally,

$$|A(z)| = \frac{4\sqrt{2}}{3} z^{3/2}, \quad \|F_{A(z)} - F_{A(z')}\|_{\infty} \leq 2\sqrt{2Z}|z - z'| \quad (0 \leq z, z' \leq Z).$$

The same modulus holds for v_H . Applying (3.4) on a finite depth mesh and using this modulus proves

Proposition 1. *For each $Z < \infty$,*

$$\varepsilon_R(Z) := \sup_{\varphi, 0 \leq z \leq Z} |v_R(z, \varphi) - v_H(z)| \longrightarrow 0. \quad (3.5)$$

4 A uniform integrable bound at all depths

We give the uniform curvature-normalized form of the method in [2, Theorem 5.1]. Let $r = 1/\kappa_+$. The support function $H(\varphi) = h_K(u_{\varphi})$ is C^2 and satisfies $H + H'' = 1/\kappa_{\varphi}$. This follows from the C^1 inverse Gauss map and $p'_K(\varphi) = \tau_{\varphi}/\kappa_{\varphi}$.

Lemma 2. *For every φ , $\overline{B}(p_K(\varphi) - ru_{\varphi}, r) \subset K$.*

Proof. Fix φ_0 , set $c = p_K(\varphi_0) - ru_{\varphi_0}$ and

$$F(t) = H(\varphi_0 + t) - c \cdot u_{\varphi_0+t} - r.$$

Then $F(0) = F'(0) = 0$ and $F'' + F = \kappa_{\varphi_0+t}^{-1} - r \geq 0$. For $|t| \leq \pi$,

$$F(t) = \int_0^t (\kappa_{\varphi_0+v}^{-1} - r) \sin(t - v) dv \geq 0.$$

For negative t both the sine and integration orientation change sign. These angles cover all support directions, proving inclusion by the support-function criterion. $\square$

Suppose $R \geq R_0 := \sqrt{2}/r$. At physical depth $0 < h \leq \sqrt{2}$, the tangent disk of radius Rr gives

$$|A_{R,\varphi}(z)| \geq \int_0^h 2\sqrt{2Rry - y^2} dy \geq \frac{4}{3}\sqrt{Rr} h^{3/2} \geq b_K z^{3/2}, \quad (4.1)$$

$$b_K = \frac{4}{3}\sqrt{r\kappa_-} > 0.$$

For $h \geq \sqrt{2}$, the cap contains the disk of radius $\sqrt{2}/2$ centered at $p_R - (\sqrt{2}/2)u_{\varphi}$. This disk is inside the tangent disk and meets every rotated, translated square lattice, by its covering radius. Hence

$$v_R(z, \varphi) = 0 \quad \text{when } q_{R,\varphi}^{1/3} z \geq \sqrt{2}. \quad (4.2)$$

The avoidance estimate [2, Lemma 3.2], [3, Theorem 1.1], supplies dimension-only constants $\nu, c > 0$ with

$$\mathbb{P}\{L \cap B = \emptyset\} \leq c/|B| \quad \text{for every convex body } B \text{ with } |B| \geq \nu. \quad (4.3)$$

We apply it to the deterministic *physical* cap, with the original randomized square-lattice law, not to an anisotropically normalized lattice. If $b_K z^{3/2} < \nu$, the trivial bound suffices; otherwise (4.1) and (4.3) apply. With $A_K = \max(\nu, c)/b_K$, this proves

Proposition 2. *For all $R \geq R_0$, $z > 0$ and φ ,*

$$v_R(z, \varphi) \leq \min(1, A_K z^{-3/2}). \quad (4.4)$$

The constants ν, c are not evaluated numerically here. This bound establishes integrability, not a numerical approximation error in R .

5 The complete expectation and the curvature factor

Pointwise convergence transfers (4.4) to v_H . In particular,

$$\int_Z^\infty v_R(z, \varphi) dz \leq \frac{2A_K}{\sqrt{Z}}, \quad \int_Z^\infty v_H(z) dz \leq \frac{2A_K}{\sqrt{Z}}, \quad Z > 0. \quad (5.1)$$

Thus $C_{\text{void}} = \int_0^\infty v_H(z) dz < \infty$. Haar-affine intensity is one. The Markov argument of [2, Lemma 3.3], applied here directly through translation intensity, gives

$$v_H(z) \geq (1 - |A(z)|)_+, \quad C_{\text{void}} \geq \frac{3}{5} \left(\frac{3}{4\sqrt{2}} \right)^{2/3} > 0. \quad (5.2)$$

Moreover,

$$\sup_\varphi \int_0^\infty |v_R(z, \varphi) - v_H(z)| dz \leq Z\varepsilon_R(Z) + \frac{4A_K}{\sqrt{Z}} \longrightarrow 0, \quad (5.3)$$

where the limit means first $R \rightarrow \infty$ at fixed Z , then $Z \rightarrow \infty$.

For $R \geq R_0$, RK contains a disk of radius at least $\sqrt{2}$. Every such disk contains the four vertices of a square-lattice cell, since each is at most $\sqrt{2}$ from its center. Thus Q_R has dimension two for every lattice in (1.1). Cauchy's perimeter formula, the support-gap survival identity and Tonelli yield

$$\mathbb{E}[P(RK) - P(Q_R)] = \int_0^{2\pi} q_{R,\varphi}^{1/3} \int_0^\infty v_R(z, \varphi) dz d\varphi. \quad (5.4)$$

This is also the planar form of [2, Lemma 3.1]; $P = \pi W$ under their mean-width convention. Integrating (5.3), with $d\varphi = \kappa ds$, gives

$$\lim_{R \rightarrow \infty} R^{1/3} \mathbb{E}[P(RK) - P(Q_R)] = C_{\text{void}} \int_0^{2\pi} \kappa_\varphi^{1/3} d\varphi = C_{\text{void}} \int_{\partial K} \kappa^{4/3} ds. \quad (5.5)$$

This proves the leading asymptotic apart from the row-model identification and numerical evaluation of the constant, which follow below.

6 Stationary parabolic support deficit

For a Haar-affine lattice Λ , define

$$D_\Lambda(t) = \min_{(x,y) \in \Lambda \cap \mathcal{P}} (y - tx + t^2/2) \geq 0. \quad (6.1)$$

The minimum exists. A lattice has a finite covering radius, and $\mathcal{P}$ contains arbitrarily large disks. A bounded value of the cost bounds x through $(x - t)^2/2$, and then bounds y . A single admissible candidate bounds the minimum uniformly on compact t -intervals, leaving finitely many possible minimizers there.

The lower boundary of $\text{conv}(\Lambda \cap \mathcal{P})$ is a locally finite polygonal chain defined over the whole real axis. Covering-radius arguments give admissible points with arbitrarily large positive and negative abscissae. On a compact abscissa interval, two bracketing points bound the lower graph height, so its vertices lie in a compact set and are finite in number.

The determinant-one affine transformations

$$S_b(x, y) = (x + b, y + bx + b^2/2)$$

preserve $\mathcal{P}$ and satisfy $D_{S_b\Lambda}(t) = D_\Lambda(t - b)$. Haar-affine invariance makes this process stationary. At $t = 0$, its survival function is v_H , with cap boundaries null under the translation average; therefore

$$\mathbb{E}_{\mu_{\text{aff}}} D_\Lambda(0) = C_{\text{void}}. \quad (6.2)$$

Let e be a maximal lower edge with endpoints $(X_L, Y_L), (X_R, Y_R)$ and slope t_e , where $X_L < X_R$. After S_{-t_e} it is horizontal at height $d = D_\Lambda(t_e)$. Its lattice points form a primitive arithmetic progression with horizontal step $a > 0$, and its extremes $x_L = X_L - t_e$, $x_R = X_R - t_e$ are those in $[-\sqrt{2d}, \sqrt{2d}]$. If $x_L > 0$, its predecessor $x_L - a$ would lie in this window: indeed $x_R \geq x_L + a$ and the window is symmetric. This contradicts extremality. The other side is analogous. Consequently

$$x_L \leq 0 \leq x_R, \quad X_L \leq t_e \leq X_R. \quad (6.3)$$

For adjacent edges, this places their slopes on the appropriate sides of $[X_L, X_R]$. The minimizing support vertex is the left endpoint on $[X_L, t_e]$ and the right endpoint on $[t_e, X_R]$. Direct integration gives

$$\begin{aligned} \int_{X_L}^{X_R} D_\Lambda(t) dt &= \int_{x_L}^0 (d - ux_L + u^2/2) du + \int_0^{x_R} (d - ux_R + u^2/2) du \\ &= d(x_R - x_L) - \frac{x_R^3 - x_L^3}{3} =: c_e \geq 0. \end{aligned} \quad (6.4)$$

The cubic coefficient is $1/3$ for this support deficit; the geometric area between the chain and parabola is a different observable.

7 Nonnegative mass transport and primitive marked Haar unfolding

The projected intervals $[X_L(e), X_R(e))$ partition $\mathbb{R}$ up to endpoints. Define

$$T_\Lambda(B, C) = \sum_e \int_{B \cap [X_L(e), X_R(e))} D_\Lambda(t) \mathbf{1}_C(t_e) dt.$$

Its source is $D_\Lambda(t) dt$; its destination at t_e carries c_e . Under S_b , both coordinates shift by b . With $I_n = [n, n + 1)$, diagonal translation invariance and Tonelli give

$$\mathbb{E}T(I_0, \mathbb{R}) = \sum_n \mathbb{E}T(I_0, I_n) = \sum_n \mathbb{E}T(I_{-n}, I_0) = \mathbb{E}T(\mathbb{R}, I_0).$$

Using (6.2),

$$C_{\text{void}} = \mathbb{E} \sum_{t_e \in [0, 1)} c_e. \quad (7.1)$$

All terms are initially nonnegative extended integrals; finiteness follows from Section 5. No signed cancellation or boundary-flux estimate is needed.

Mark each maximal edge by its unique primitive lattice vector pointing right. Several collinear points give one maximal edge, not one edge per adjacent pair. For a fixed marked direction, the first occupied row after shear gives an edge precisely when it has at least two points; a singleton contributes zero.

Write the marked vector as (a, b) , $a > 0$. A representative oriented lattice basis is

$$M = \begin{pmatrix} a & b \\ \beta & b\beta/a + 1/a \end{pmatrix}, \quad 0 \leq \beta < a.$$

The primitive marked Haar measure is

$$\frac{1}{\zeta(2)} da db \frac{d\beta}{a}. \quad (7.2)$$

To specify the unfolding: primitive integer vectors form the orbit of e_1 under $\mathrm{SL}(2, \mathbb{Z})$. Its stabilizer consists of $\begin{pmatrix} 1 & 0 \\ n & 1 \end{pmatrix}$, $n \in \mathbb{Z}$. Unfolding a nonnegative primitive sum replaces the lattice fundamental domain by the stabilizer quotient. In [1, Section 7, Lemma 7.4 and (7.17)], the continuous stabilizer has coordinate u and Haar measure du ; quotienting gives $0 \leq u < 1$. Here $u = \beta/a$. The coefficient is $1/\zeta(2)$. This gives the full marked law, including shear; the primitive Siegel marginal alone would not specify that conditional law.

Put $b = at$, so $da db = a da dt$. The shear S_{-t} makes the basis $(a, 0), (\beta, 1/a)$. The torus translation is uniform on the area-one rectangle

$$0 \leq \xi < a, \quad 0 \leq \delta < 1/a,$$

obtained by reducing height, then abscissa. Conditional on (a, t) , the phases have joint probability measure $(d\beta/a) d\xi d\delta$: they are independent uniform variables on their respective intervals.

8 The row model and its exact formulas

For $a > 0$ choose independent $\beta, \xi \in [0, a)$ and $\delta \in [0, 1/a)$ uniformly. The row points are

$$(\xi + aj + k\beta, \delta + k/a), \quad j, k \in \mathbb{Z}. \quad (8.1)$$

Rows $k < 0$ do not meet $\mathcal{P}$. Let k_* be the first occupied row in $\mathcal{P}$, $d = \delta + k_*/a$ and $x_1 \leq x_2$ its extremes in $[-\sqrt{2d}, \sqrt{2d}]$. Set

$$c = d(x_2 - x_1) - (x_2^3 - x_1^3)/3, \quad G(a) = \mathbb{E}c. \quad (8.2)$$

The first row exists since every window of length at least a contains a point. A singleton contributes zero. Applying (7.2) to (7.1), with the stated phase law, gives

$$C_{\text{void}} = \frac{1}{\zeta(2)} \int_0^\infty \int_0^1 a G(a) dt da = \frac{6}{\pi^2} \int_0^\infty a G(a) da. \quad (8.3)$$

There is no additional factor $1/2$: $a > 0$ orients every edge once. The identity also proves integrability of $aG(a)$.

8.1 A phase average valid for any number of row points

Put $r = \sqrt{2d}$. If $2r \geq a$, the rightmost row point is uniform on $(r - a, r]$, and the leftmost on $[-r, -r + a)$. These are marginal laws, requiring no independence of the extremes and holding even with three or more points. The odd function $F_d(x) = dx - x^3/3$ gives

$$\mathbb{E}c = \frac{2}{a} \int_{r-a}^r (r^2x/2 - x^3/3) dx = \frac{(2r-a)(r^2 + 2ar - a^2)}{6} =: \mathcal{H}(d, a). \quad (8.4)$$

If $2r < a$, an occupied row is a singleton. This proves the phase formula without relying on an earlier heuristic model note.

For $0 < a \leq 2$, row 1 always has a window of length at least a . The phases of rows 0 and 1 are independent uniform modulo a by an integer torus change of variables of determinant one. Thus $G = G_0 + G_1$, with

$$\begin{aligned} G_0(a) &= a \int_{a^2/8}^{1/a} \mathcal{H}(d, a) dd \\ &= \frac{4\sqrt{2}}{15} a^{-3/2} + \frac{1}{2} - \frac{4\sqrt{2}}{9} a^{3/2} + \frac{a^3}{6} - \frac{17a^6}{5760}, \\ G_1(a) &= a \int_0^{a^2/8} \left(1 - \frac{2\sqrt{2d}}{a}\right) \mathcal{H}(d + 1/a, a) dd. \end{aligned} \quad (8.5)$$

The first integral accounts for row 0 when its contribution can be nonzero; the second accounts for its being empty and row 1 being first occupied. Formula (8.4) covers any point multiplicity in these rows.

8.2 Exact treatment of all rows when $a \geq 2$

Let $n = a^3/8 \geq 1$, $x = a\delta \in [0, 1)$ and $k_0 = \lceil n - x \rceil$. This is the first row whose window length is at least a . Earlier rows have at most one point, and this row has at most two. All later rows are nonempty. A nonzero first-row contribution therefore requires two points at k_0 and every earlier row empty.

Write $D = x + k_0 = n + f$, $0 \leq f < 1$, and normalize phases and abscissae by a . Define

$$R_f = \frac{1}{2}\sqrt{D/n}, \quad e = R_f - \frac{1}{2}, \quad t_j = \frac{1}{2} - \frac{1}{2}\sqrt{(D-j)/n} \quad (1 \leq j \leq k_0).$$

Two points in the final row correspond to centered phase $z \in [-e, e]$. An empty preceding row corresponds to $z_p \in (-t_1, t_1)$. These two phases are independent uniform: the change from initial phase and shear to the consecutive row phases has determinant -1 on the torus. Take the lift $b = z - z_p$ of the normalized shear; $|b| \leq e + t_1 = R_f - R_{f-1} \leq 1/2$. The phase j rows earlier lifts to $z_j = jz_p - (j-1)z$. Concavity gives

$$\sqrt{D-j} \leq j\sqrt{D-1} - (j-1)\sqrt{D}, \quad t_j \geq jt_1 + (j-1)e.$$

Hence $|z_j| < jt_1 + (j-1)e \leq t_j \leq 1/2$, proving all earlier rows empty with no missed wrap modulo one. Conversely their emptiness implies emptiness of the preceding row. Boundary equalities have zero phase measure.

The final height $d = D/a$ is uniform on $[h, h + 1/a)$, $h = a^2/8$, since $f = (x - n) \pmod{1}$ is uniform. The preceding-row emptiness probability is $1 - 2\sqrt{2(d - 1/a)}/a$. Its phase is independent of the final phase, giving the exact formula

$$G(a) = a \int_h^{h+1/a} \left(1 - \frac{2\sqrt{2(d - 1/a)}}{a}\right) \mathcal{H}(d, a) dd, \quad a \geq 2. \quad (8.6)$$

Possible first-occupied singletons before this full window remain genuine events but have zero contribution, which is why no further row case enters this integral.

9 Compact integral representations and a rational certificate

Decompose $J = \int_0^\infty aG(a) da = J_0 + J_1 + J_T$ according to the two contributions in (8.5) and the tail (8.6). Direct integration gives

$$J_0 = \frac{32}{15} + 1 - \frac{128}{63} - \frac{17}{180} = \frac{141}{140}.$$

For J_1 , substitute $a = 2v^2$ and $d = (a^2/8)t^2$. With $W = \sqrt{1 + v^6t^2}$,

$$J_1 = \frac{16}{3} \int_0^1 \int_0^1 v^6 t(1-t)(W - v^3)(W^2 + 4v^3W - 4v^6) dt dv. \quad (9.1)$$

For J_T , substitute $a = 2/v$ and rationalize the square-root differences:

$$J_T = \frac{4}{3} \int_0^1 \int_0^1 \frac{f(1-f)[v^3f - 3 + 4\sqrt{1 + v^3f}]}{(1 + \sqrt{1 - v^3(1-f)})(1 + \sqrt{1 + v^3f})} df dv. \quad (9.2)$$

Both integrands are continuous on the compact square. The apparent endpoint issue of the second formula has a positive denominator and a vanishing numerator factor.

9.1 A refinement of the model tail

With $u = 8/a^3$ and $d = h + f/a$, formula (8.6) becomes

$$G(a) = \frac{a^3}{24} \int_0^1 [1 - \sqrt{1 - u(1-f)}][(uf - 7)\sqrt{1 + uf} + 7 + 3uf] df.$$

The product expands uniformly as

$$\frac{u^2 f(1-f)}{4} + \frac{u^3 f(1-f)(1+10f)}{16} + O(u^4).$$

Its integral is $u^2/24 + u^3/16 + O(u^4)$, whence

$$G(a) = \frac{1}{9a^3} + \frac{4}{3a^6} + O(a^{-9}), \quad a^3 G(a) = \frac{1 + 12/a^3}{9} + O(a^{-6}). \quad (9.3)$$

This corrects the former relative a^{-2} heuristic. It is a statement about the row-model tail, not a second-order physical asymptotic in R .

9.2 Exact series and truncation errors

Let $b_n = \binom{1/2}{n}$ and $q_n = \binom{3/2}{n}$. The identity

$$(W - v^3)(W^2 + 4v^3W - 4v^6) = W^3 + 3v^3W^2 - 8v^6W + 4v^9$$

and termwise integration yield

$$J_1 = \frac{16}{3} \left[\frac{1}{20} + \frac{3}{320} + \frac{1}{24} + \sum_{n=0}^{\infty} \frac{q_n/(6n+7) - 8b_n/(6n+13)}{(2n+2)(2n+3)} \right]. \quad (9.4)$$

For $N \geq 2$, the alternating binomial remainders bound the error of truncation at $n = N$ by

$$E_1(N) = \frac{16}{3(2N+4)(2N+5)} \left(\frac{|q_{N+1}|}{6N+13} + \frac{8|b_{N+1}|}{6N+19} \right). \quad (9.5)$$

For the tail, write $A(x) = 1 - \sqrt{1-x} = \sum_{m \geq 1} \alpha_m x^m$ and

$$P(y) = (y-7)\sqrt{1+y} + 7 + 3y = \sum_{n \geq 1} p_n y^n,$$

$$\alpha_m = -(-1)^m b_m > 0, \quad p_1 = \frac{1}{2}, \quad p_n = b_{n-1} - 7b_n \quad (n \geq 2).$$

The absolutely convergent integrated series is

$$J_T = \frac{4}{3} \sum_{m,n \geq 1} \alpha_m p_n \frac{m!n!}{(m+n+1)![3(m+n)-5]}. \quad (9.6)$$

Indeed $\sum \alpha_m = 1$ and $\sum |p_n| = 5$, so Tonelli for absolute values justifies the exchanges. Put $T_N = \binom{2N}{N}/4^N$. The omitted m -tail has total coefficient mass T_N ; the absolute n -tail has mass $T_{N-1} + 7T_N$. For an omitted index at least $N+1$, its beta-integral weight is at most $1/[(N+2)(N+3)]$ and its remaining denominator at least $3N+1$. Adding the two possibly overlapping positive bounds gives

$$E_T(N) = \frac{4}{3} \frac{T_{N-1} + 12T_N}{(N+2)(N+3)(3N+1)}. \quad (9.7)$$

The certificate evaluates the finite sums entirely as fractions, adds (9.5) and (9.7), and encloses π using

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$$

with 30 terms and alternating rational remainders in each arctangent. From $J \in [J_-, J_+]$ and $0 < \pi_- \leq \pi \leq \pi_+$,

$$\frac{6J_-}{\pi_+^2} \leq C_{\text{void}} \leq \frac{6J_+}{\pi_-^2}.$$

At $N = 100$, $E_1 < 4.607 \cdot 10^{-10}$ and $E_T < 3.090 \cdot 10^{-7}$. Exact outward decimal conversion gives

$$\boxed{0.719232985957311632 < C_{\text{void}} < 0.719233362190182706}. \quad (9.8)$$

The weaker enclosure in Theorem 1 follows. Rational comparisons with the rounding thresholds certify the rounded values 0.719 and 0.719233. Neither is asserted to be the exact constant. The certificate verifies the evaluation of the formulas; their model identification is the mathematical argument in Sections 6–8.

For orientation only, floating quadratures give

$$J_1 \simeq 0.100019051910680863, \quad J_T \simeq 0.0759292423148140273, \\ C_{\text{void}} \simeq 0.719233174880505782.$$

These extra displayed digits have no certified significance beyond (9.8).

10 Scope, open questions and provenance

Theorem 1 is now proved under its stated assumptions by combining (5.5), (8.3) and (9.8). Disks and ellipses are covered. The usual superellipses with $p > 2$ have zero curvature at the axes, and those with $1 < p < 2$ are not C^2 there; neither class is included. No second-order coefficient, convergence rate in R , deterministic unrotated-lattice analogue, or implication for the Riemann hypothesis is obtained here.

The manuscript concerns a fixed K ; uniformity over a family would require shared geometric and continuity bounds. The tail refinement (9.3) does not resolve the empirical second term. Spectral suggestions discussed in [9] remain exploratory and are not used in any proof here.

AI contributions and responsibility. Patrick Royer formulated and directed the research questions and remains the responsible human author and submitter. ChatGPT/Codex contributed mathematical development, implementation, assisted validation, manuscript drafting and final integration. Claude supplied an external AI audit and a separately written verification script; their limitations and the outward-rounding correction are documented. Earlier exploratory inputs supplied by the author included DeepSeek notes; their precise contribution to the present proofs has not been independently reconstructed. AI systems are acknowledged as tools, not as human authors, peer reviewers or scientific guarantors. No independent human mathematical review or formal proof-assistant verification is attested.

Software and rights. The rational certificate uses Python’s standard library. Diagnostic calculations use NumPy, SciPy, mpmath and SymPy; composition and rendering use L^AT_EX and Poppler. These are freely available tools; this statement does not assert that the AI services used were free of charge. The supplied DEL 1.1 licence and the accompanying scope notice govern the corpus to the extent of the author’s rights. Third-party software and references retain their own terms. The licence grants no exclusive right over mathematical formulas or factual results.

A Published inputs in the specialized form used here

These are mathematical restatements, with the conventions of this manuscript, rather than reproductions of the original wording.

A.1 Expanding circles

The $d = 2$, level-one specialization of [1, Corollary 5.9] is (3.4), for bounded continuous F on X_0 and normalized absolutely continuous angular measure. In our application the measure is uniform, $M = I$, and the family of tests is constant. Continuity is proved by Lemma 1; the affine extension is obtained by exact torus averaging. This is why no equidistribution theorem for an unbounded cusp observable is inserted into the cap-convergence proof.

A.2 Empty-cap identity and avoidance

The planar specialization of [2, Lemma 3.1], under $P = \pi W$, is the physical-depth identity

$$\mathbb{E}[P(RK) - P(Q_R)] = \int_0^{2\pi} \int_0^\infty \mathbb{P}\{(RK)_{h,\varphi} \cap L = \emptyset\} dh d\varphi.$$

It was also derived directly in Section 5. The $d = 2$ avoidance input of [2, Lemma 3.2] is (4.3), with dimension-only constants and the law (1.1); its cited primary source is [3, Theorem 1.1], which credits the planar case to [4].

A.3 Marked Haar measure

The stabilizer disintegration of [1, Lemma 7.4, (7.17)], combined with the primitive orbit unfolding and stabilizer quotient, gives (7.2). The argument retaining the conditional shear and uniform torus translation is in Section 7. This is a marked counting formula, not just an unmarked mean-value assertion.

B Reproducibility and verification status

The publication archive contains the L^AT_EX source, the standard-library rational certificate and exact fractions, the corrected independent verification script and its reference outputs, source proof notes and audit provenance, licence scope, citation metadata and checksums. The minimal numerical command is

```
python code/cert_identity_rational.py -N 100 -output results/CERTIFICATE.json
```

The two independent rational implementations use slightly different valid arctangent enclosures, yet yield the same outward 18-digit display. The complete Claude script replay checks symbolic identities, direct and substituted quadratures, nine Monte Carlo parameters with two million samples each, and model-tail diagnostics. Its original nearest-rounded C2 display is preserved as historical evidence; the corrected copy alone is the reference for its outward decimal bounds.

Item	Status and interpretation
Cap convergence / complete expectation	Mathematical derivation in Sections 2–5, relying on cited published inputs.
Constant identification	Stationarity, mass transport and marked unfolding in Sections 6–8.
Numerical enclosure	Exact fraction computation with proved truncation bounds, Section 9.
Monte Carlo / quadratures	Diagnostics, not proof or rigorous quadrature error bounds.
AI audits	Assisted checks; no independent human peer review attested.
Publication	This is a prepared preprint; new public DOI and release must be recorded after deposition.

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